Exercise 2:

a) We need to prove:

① ($\Omega \in \mathcal{F}$): Since $\Omega^c = \emptyset$ is at most countable, then, by definition of $\mathcal{F}$, $\Omega \in \mathcal{F}$.

② ($F \in \mathcal{F} \Rightarrow F^c \in \mathcal{F}$): Let $F \in \mathcal{F}$, then F or F^c is at most countable. If F^c is at most countable, then, by dedinition of $\mathcal{F}$, $F^c \in \mathcal{F}$. If F^c is not countable, then F must be countable. Since $(F^c)^c = F$ is countable, F^c satisfy the condition to be in $\mathcal{F}$.

③ ($F_n \in \mathcal{F}\ \forall n \in \mathbb{N} \Rightarrow \bigcup_{n\in\mathbb{N}} F_n \in \mathcal{F}$): Let $F_n \in \mathcal{F}\ \forall n \in \mathbb{N}$. We distinguish two cases:

(1) F_n is at most countable $\forall n \in \mathbb{N}$: Then, $\bigcup_{n\in\mathbb{N}} F_n$ is at most countable and, therefore, $\bigcup_{n\in\mathbb{N}} F_n \in \mathcal{F}$.

(2) At least one $F_k \in \{F_1, F_2, \dots\}$ is not countable. Then, F_k^c must be at most countable (by definition of $\mathcal{F}$). Since $(\bigcup_{n\in\mathbb{N}} F_n)^c = \bigcap_{n\in\mathbb{N}} F_n^c \subseteq F_k^c$, then $(\bigcup_{n\in\mathbb{N}} F_n)^c$ is also at most countable, and $\bigcup_{n\in\mathbb{N}} F_n \in \mathcal{F}$ □

b) We need to show:

① ($\sigma(\mathcal{G}) \subseteq \mathcal{F}$): For any $G = \{\omega\} \in \mathcal{G}$, $G \in \mathcal{F}$ because singletons are countable and countable sets are elements of $\mathcal{F}$. Since $\mathcal{F}$ is σ-algebra and $\sigma(\mathcal{G})$ is the smalles σ-algebra that contains $\mathcal{G}$, then $\sigma(\mathcal{G}) \subseteq \mathcal{F}$.

② ($\mathcal{F} \subseteq \sigma(\mathcal{G})$): Let $F \in \mathcal{F}$, then F or F^c are at most countable:

- If F is at most countable, then it can be expressed as a sequence $(f_n)_{n\in I}$ where $I \in \mathbb{N}$. Therefore, $F = \bigcup_{n\in I} \{f_n\}$. Since $\{f_n\} \in \mathcal{G} \subseteq \sigma(\mathcal{G})\ \forall n \in I$ and $\sigma(\mathcal{G})$ is σ-algebra, then $F \in \sigma(\mathcal{G})$.
- If F is not countable, then F^c is at most countable and, as before, $F^c (= \bigcup_{n\in I} \{f_n^*\}, I \in \mathbb{N}, f_n^* \in F^c) \in \sigma(\mathcal{G})$ and, therefore, $(F^c)^c = F \in \sigma(\mathcal{G})$ as $\sigma(\mathcal{G})$ is σ-algebra. □

c) We need to prove:

① ($P(\Omega)=1$): Since $\Omega^c=\emptyset$ is at most countable, then $\mathbb{P}(\Omega)=1$.

② (σ-additivity): Let $(F_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ s.t. $F_k \cap F_\ell = \emptyset$ $\forall k\neq \ell$. We distinguish two cases:

(i) If F_n is at most countable $\forall n\in\mathbb{N}$, then $\mathbb{P}(F_n)=0$ $\forall n\in\mathbb{N}$ and $\sum_{n\in\mathbb{N}} \mathbb{P}(F_n)=0$.
Also, $\bigcup_{n\in\mathbb{N}} F_n$ is at most countable and, then, $\mathbb{P}(\bigcup_{n\in\mathbb{N}} F_n)=0$.
Therefore, $\mathbb{P}(\bigcup_{n\in\mathbb{N}} F_n) = \sum_{n\in\mathbb{N}} \mathbb{P}(F_n)$

(ii) If there exist $k\in\mathbb{N}$ s.t. F_k is uncountable. By definition of $\mathcal{F}$, its complement F_k^c is at most countable, and, then, $\mathbb{P}(F_k)=1$. Since $F_k\cap F_\ell=\emptyset$ $\forall k\neq\ell$, then

$F_\ell \subseteq F_k^c$ $\forall \ell\neq k$. Thus, F_ℓ is at most countable (*) and $P(F_\ell)=0$ $\forall \ell\neq k$. Therefore:

$$\sum_{n\in\mathbb{N}} \mathbb{P}(F_n) = \mathbb{P}(F_k) + \sum_{\ell\neq k} \mathbb{P}(F_\ell) = 1+0=1$$

Since F_k^c is at most countable and $\left(\bigcup_{n\in\mathbb{N}} F_n\right)^c = \bigcap_{n\in\mathbb{N}} F_n^c \subseteq F_k^c$, then $\left(\bigcup_{n\in\mathbb{N}} F_n\right)^c$ is at most countable and $\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} F_n\right)=1$.

Therefore, $\mathbb{P}(\bigcup_{n\in\mathbb{N}} F_n) = \sum_{n\in\mathbb{N}} \mathbb{P}(F_n)$ $\square$.